

The Power of Proxmox Backup Server with Proxmox VE

The Proxmox Backup Server is an open source backup software project that supports virtual machines, containers, and physical hosts. Based on the Debian Linux distribution, it comes with out-of-the-box ZFS support. We look at its key features and benefits.



The Proxmox Backup Server (PBS) is a powerful, enterprise-grade backup solution designed to integrate seamlessly with the Proxmox Virtual Environment (VE). This integration offers numerous benefits, from efficient data backup to advanced security features, making it an ideal choice for IT managers already using the standalone ProxmoxVE to cluster, which has many nodes of ProxmoxVE. There are quite a few reasons for using PBS. Some of these are discussed here.

Seamless integration with Proxmox VE

Proxmox Backup Server is fully integrated with Proxmox VE, ensuring a streamlined and cohesive experience for managing virtual machine (VM) and container backups. This integration allows users to easily add PBS datastores and run backups just like any other Proxmox VE storage type. The tight integration also supports features such as live-restore, which allows VMs to start running as soon as the restore process begins, minimising downtime and enhancing productivity.

Efficient incremental and deduplicated backups

One of the standout features of Proxmox Backup Server is its support for incremental backups and deduplication. Incremental backups only store changes made since the last backup, significantly reducing storage requirements and network load. Deduplication further optimises storage usage by eliminating

duplicate data across backups. This efficiency is achieved through the Proxmox File Archive Format (.pxar) and QEMU dirty bitmaps, ensuring fast and storage-efficient backups.

Fast and space-efficient compression

Proxmox Backup Server employs Zstandard compression, capable of compressing several gigabytes of data per second. This high-speed compression ensures that backups are not only quick but also space-efficient, saving valuable storage resources. The combination of incremental backups, deduplication, and efficient compression makes PBS a robust solution for managing large volumes of data with minimal storage overhead.

Enhanced data security

Data security is paramount in any IT environment, and Proxmox Backup Server excels in this area with strong client-side encryption using AES-256 in Galois/Counter Mode (GCM). This encryption ensures that data remains secure before it even reaches the backup server, making it useless for unauthorised persons. Additionally, PBS supports the use of master keys and secure storage of encryption keys, adding another layer of protection for sensitive data.

Ransomware protection

PBS has several features designed to protect against ransomware. Off-server with tape backup and offsite remote synchronisation capabilities enable recovery even in the event of a ransomware attack. Moreover, the system uses SHA-256 checksums and verification jobs to detect potential ransomware threats. By ensuring that existing blocks of data are not rewritten, PBS minimises the risk of data corruption and loss due to ransomware.

Quick and granular restore options

The server offers fast and straightforward restore options through its intuitive GUI. Users can quickly search for and restore individual files, directories, or entire archives. The ability to download restored data as compressed archives further enhances the flexibility and convenience of the restore process. These granular restore options make it easy to